

The design and analysis of an advanced hybrid renewable energy microgrid for Cyangugu Remote rural community in Rwanda using Homer Pro

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ABSTRACT: In Rwanda, and in Cyangugu Remote community near the Kivu Lake, there is high solar sunshine energy, significant wind energy, abundant methane gas, appreciable geothermal energy, peat(biomass), hydro renewable energy sources, and grid power access. To ensure feasibility, accessibility, and sustainability, a balance need to be maintained to optimize this energy mix options, in terms of cost (NPV, COE etc.), maintenance, sensitivity, optimization, and other system parameters. The best combination of these Hybrid Renewable Energy Sources (HRES) considering the community's energy needs, is modelled using the Hybrid Optimization of Multiple Electric Renewables – Professional (HOMER Pro), innovatively, to design and find out the proposed optimum system requirements with environmental advantages or benefits, best payback duration, lower or zero net GHGs and radiation emissions, performance optimization with lest value of Net Present Cost (NPC) and Cost of Energy (COE), sensitivity analysis involving components supply changes like wind speed change based atmospheric conditions, variable sunshine frequencies based on time of the day, seasons, fuel price changes etc. These seven energy supply components is used to provide access to modern, clean and reliable energy to Cyangugu.

Keywords: hydrogen, renewable, storage, hybrid, energy mix, microgrid, modular: inverter, power quality, photo voltaic array, natural gas, biomass, peat

1 INTRODUCTION

Cyangugu is a small town in Rwanda, and a border town with the Democratic Republic of Congo (Congo DRC). Due to its proximity to lake Kivu, that has large natural gas deposit, it has large supply of methane gas. The hill topography of the environments present good hydro energy resources, the abundant sunshine ensures sufficient solar energy resources, the peat (decayed vegetation/organic matter accumulation in natural areas called peatlands), significant wind energy resources, Power grid access point [1,2], a standalone microgrid system will help provide clean and modern renewable energy to this remote tourist community separated from the Congo DR by the Kivu Lake. The microgrid analysis is done using homer pro, which needs only basic parameters like the load requirements, cost of components, etc. to perform complete power system and economic analysis of the microgrid. A Hybrid Renewable Energy System (HRES) is shown in Figure 1, the Optimization of Multiple Electric Renewables – Professional (HOMER Pro), innovatively and systematically evaluates and analyse off grid and grid connected power systems in a way that reduces the time and human effort required, when designing a power system or an advanced microgrid consisting of seven energy supply components.

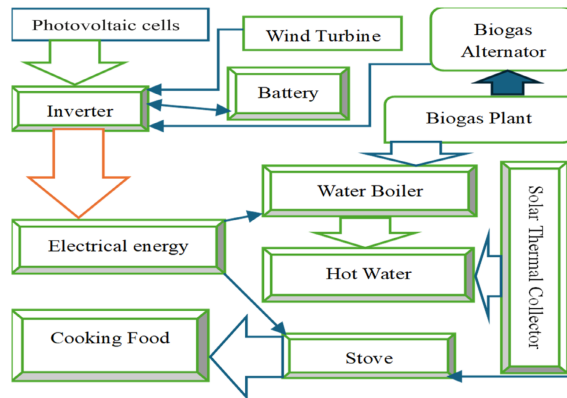


Figure 1. A hybrid renewable energy system (HRES) [3].

This helps in decisions making about the system configurations, selection of the best components, the size and number of components required, sensitivity analysis, most efficient components, cost considerations, optimum system conditions, technology, and energy renewable resources availability are also provided to make selection of the best system configuration easier [4].

2 PROBLEM STATEMENT

The About 80% of Rwandans are living in the rural communities [5], energy access through the national grid is limited, thus, there is a need for analyses of an advanced microgrid with access to the national grid that can supply a remote community sustainably and reliably. The potential geothermal renewable energy resource (temperature from 90°C to 140°C), and the geothermal fluid mass flow (from 20 kg/s to 80 kg/s) [6], the windier conditions from May to September each year [7], the solar energy from sufficient sunshine, and the river flow due to the gradients created by the hills and landscape of the Cyangugu community can provide renewable energy from an advanced microgrid utilizing this distributed generation.

3 LOAD DEMAND ASSESSMENT

The population of Cyangugu community is 63,883 persons [8,9], which translates to 12828 households of five persons, government health expenditure per person is 3 015,00 Rwandan franc. Using the minimum threshold electricity consumption level of 250 kWh/annum for a rural household, the Cyangugu community energy need is 3206900kWh/annum or 8786kWh/day translating to (8786kWh/24h) 366.1kW minimum power plant overall capacity. If power factor and other losses are factored in, then it should be 0.5MW power plant total capacity.

4 SOCIAL RESOURCE ASSESSMENT

The renewable energy resources assessed includes solar energy (5.2GWh/m²/day), hydro-energy (0.35TWh), geothermal energy (0.01TWh), natural gas (0.22TWh), and wind energy (18.4W/m² to 24.4W/m², average = 21.4W/m², at 115.5 m², power delivered/day = 59.3208kWh/day/Wind turbine (2.5kW capacity per wind turbine) = 21652.092kWh/annum/Wind turbine) according to sustainable energy for all (SE4ALL) [10,11–13]. The average flow velocity of hydrokinetics is 6.92 m/s [14]. Biomass (peat) energy density = 15MJ/kg [15], inverter, and deep cycled Lead-Acid battery.

5 CASE STUDY DESIGN OVERVIEW

To The system configuration developed is show in Figure 2.

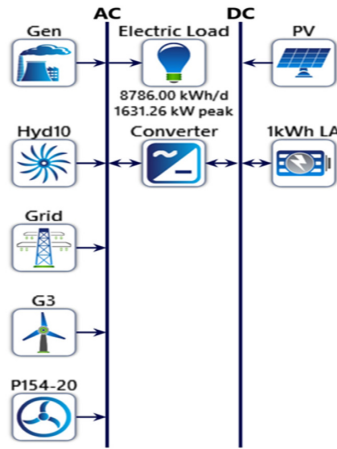


Figure 2. Advanced hybrid renewable energy scheme.

The motivation and reasons for choosing this system configuration is based on the various renewable energy resources available, accessible and the low cost associated with the installation, running, and maintaining these power supplies. The return of investment, quick payback period, low NPV, and COE, makes foreign direct investment in this energy sector [4], attractive and appealing to both local and foreign investors[16]. However, according to National Institute of Statistics of Rwanda (NISR), World Bank (WB), International Monetary Fund (IMF) and International Statistics (IS), and various data files, the discount rate is 16.2%, inflation rate is 10.78% (increasing at the rate of 2.7% per month going forward), GNI per Capital is \$850, but Rwanda has experience infrastructural and economic advancement for a decade and managed to maintain this growth despite COVID-19 pandemics [17,18]. This a pitfall for investors because increase in inflation causes decrease in Gross Domestic Product (GDP) [19], since cost of getting loan from financial institutions is high based on Rwanda financial data [10]. Because of the well-crafted sensitivity analysis, optimization analysis, and guaranteed energy resource availability, financial institution will be willing to make their loan accessible in this sector, with or without minimal commitment deposit by the investors seeking financial loan [16]. The government will incentivize this energy sector to further strengthen the investment opportunities, and the job creation associated with it, for the betterment of this community [20]. These seven components emit low GHGs, and the carbon footprint is low. The money spent on the hospital by the government and members of this community will drop, and the quality of life will certainly improve [21]. Also, to account for unmet energy load needs in Rwanda, the Annual Capacity Shortage is set at 5% (at least all load energy needs is met 95% of the time) [4].

Geographic tool, like the geographic information system (GIS) can be used to model the supply chain managements systems like the Hydrogen Supply Chain (HDC) systems. Model for Optimization of Regional Hydrogen Supply is used to do geographic analysis (and economic feasibility) of the German HDC [22]. FCEVs will be sustainable, since it is an example of hydrogen to renewable energy technology, which is reliable (it can travel long distances without the need to recharge or refuel car batteries), can be integrated into the distributed generation system, and can capture wind energy, hydro energy, nuclear energy, and solar energy directly from a natural energy rich environment.

6 MATHEMATICAL MODELLING OF SYSTEM COMPONENTS

Detailed description of each of the components used in the design are described below.

Solar energy source. A 30kWh PV generic flat array (available supply during the day) consisting of solar panel rated 300W each with an area of 1.5329m^2 is used to generate electricity at $((300\text{W} \times 12 \text{ hrs.}) / (1.5329\text{m}^2))$ $2.35\text{kWh/m}^2/\text{day}$ from the sun at 20% conversion efficiency. Solar PV contribution to the energy mix is 110kW.

Wind energy from the generic turbine (5kWh variable energy source, providing energy day and night), delivers power/day = $59.3208\text{kWh/day/each}$ standalone wind turbine at 40% conversion efficiency. The capacity and contribution to the energy mix is 2.5kW. Table 1 shows the distribution of wind speed from January to December across Rwanda.

Table 1. Wind speed across Rwanda.

Month	Wind Speed (m/s)
January	1.65
February	1.69
March	1.83
April	2.06
May	2.32
June	2.37
July	2.41
August	2.37
September	2.15
October	1.92
November	1.79
December	1.65

Hydro energy from runoff river (constant 10KWh energy supply), and hydrokinetic from runoff water sources from the hills and valleys (cheaper) as result of the Rwanda landscapes at a conversion efficiency of 90% [23]. The stream flow rate (Liter per second (L/s)) from January to December is: 280, 334, 321, 394, 327, 250, 197, 340, 378, 376, 372, 392 [15].

Storage/source. Lead-acid battery is used because of its deep cycled charging/discharging nature, and stability at various temperature variations (capacity for storage from solar PV = $120\text{AH} \times 10 = 1200\text{AH}$).

Transmission and distributions. The connecting cables is selected based on the current rating $= 300\text{W}/12\text{V} = 25\text{A}$ per solar panel connection to the inverter (each 300W capacity solar panels needs 300W inverter capacity). Losses from this component is considered in this research.

Biogas and natural (methane powered) power plant capacity is 92kW supplying 2196.5kWh/day to the energy mix.

Geothermal (conversion of earth internal heat energy to electricity) plant capacity is at 4% of total energy demand = 351.44kWh , which is 14.64kW capacity and contribution to the energy mix. The geothermal energy is used to heat water, and the hot water is a feedstock to hydrokinetic turbine, as shown in Homer Pro simulation software. It produces 20kW at 4m/s at cost of 299.43 Rwanda franc/kWh according to Guinard energies.

Peat power plant capacity = 10% of total energy requirement = 878.5kWh (3162600kJ) = 36.61kW . 0.1757 tons is required to supply this energy. The peat energy resources can last for 30 years, with heat value of 12999.69kJ/kg to 21999.47kJ/kg , (average value = 17999.57kJ/kg). [24].

National grid component energy supply is variable, considered to supply an average of 100kW to the energy mix.

Total energy mix = 100 + 94 + 37 + 15 + 10 + 110 + 2.5 = 368.5kW. I recommend an increase in supplied power from the national grid or addition of more solar panels, which will result in topping up the total energy mix value to 500KW or 5MW due losses and power quality/power factor issues.

These energy sources are optimum and best suited to form a micro grid, integrated with the national grid, that can supply this community, forming a Distributed Generation (DG).

The mathematical models for each electrical supply component as follow:

Solar PV energy density $I_s = I_b F_b + I_d F_d + F_r(I_b + I_d)$ [25]. Where I_s is the total solar radiation (kWh/m²), I_b is the direct irradiation and I_d is the diffuse solar radiation (kWh/m²), F_b is the tilt factors for the beam, F_d is the diffuse parts of solar radiation, and F_r is the reflected parts of solar radiation.

The Solar Power Generation (SPV) $PPV = R_{pv} \times I_s A$ [26]. Where PPV is the power generation from the PV array (kWh), R_{pv} is the system efficiency, A is the available panel area.

The Wind Energy (WES) $PWT = P_w \times A_w R_w$ [27]. Where PWT is the electric power obtained by the wind turbine (kWh), P_w is the wind generator power, A_w is the summation of the swept area, R_w is the total gearbox, generator, and associated components efficiency.

The Biomass energy (BM) $PBM = R_g(\varphi_w - \varphi_B)$ [28,29]. PBM is the biomass energy (kWh), R_g is the gasifier system efficiency, which includes the biomass efficiency of the producer gas conversion, internal combustion engine, and generator, φ_w is the biomass present now and instantly (kg), φ_B is the required kilogram per hour of the biomass present.

Biogas/natural gas: $PBG = R_b (BG_{total} - BGC-H)$ [30,31]. PBG is the generated power by the BG engine fired electrical machine (kWh), R_b is the biogas-fueled engine and generator system overall efficiency, BG_{total} is the biogas available instantly (m³), $BGC-H$ is the consumed biogas for clean cooking and modern generation boiling of water (m³).

Solar Thermal Collector: $PSTC = I_b R_{tc} A_a - P_{loss}$ [31]. $PSTC$ is the generated power by solar thermal collectors (kWh), R_{tc} is the optical efficiency of the collector, A_a is the absorber surface area (m²), P_{loss} is the power lost in the solar collector (kWh).

Hydro power and hydrokinetics are mathematically modelled as Off-grid operation $M = J \{(\pi/30) \times (dn/dt)\}$ [32]. Where J is the moment of inertia of the turbine, n is the rotational speed, t is the time, M is the resistance moment of the water flow.

80% of the Cyangugu community are energy poor based on national statistics and demographics, (80% of 63,883 persons) = 51107 people, which translates to 10222 households of five persons. Using the minimum threshold electricity consumption level of 250 kWh/annum for a rural household, the Cyangugu rural community energy need is 2555500kWh/annum or 7002kWh/day. To meet this energy demand, 68% of the energy need will be supplied by the solar photo-voltaic array (cheaper, and the price keeps falling), 11% from wind turbine, and 21% from hydro sources (hydrokinetics, geothermal steam turbine and hydro-dam power plants) each based on the availability and accessible renewable energy resources in the rural community. Other available sources are not currently accessible (biogas, biomass, geothermal, etc.). GHGs, climate change, and ecological issues are some constraints of peat energy source. This mathematical modelling of the load is calculated from the percentage resource availability and accessibility considering the hills (river runoff), valleys (water runoff), rainfalls (change in water volume, flow rate and water flow velocity), sunshine (night and day) and wind speeds. Thus, the total of these contributions based on these percentages will fulfill the energy demand of 43.93kWh.

Total energy = 0.68 X SGP + 0.31 X HE + 0.11 X WE. WE = Wind Energy, HE = Hydro-Energy.

7 RESULT AND DISCUSSION

The result analysis procedure which involves the project input parameters, Sensitivity analysis, optimization, and simulation is shown in Figure 3 flow chart [15].

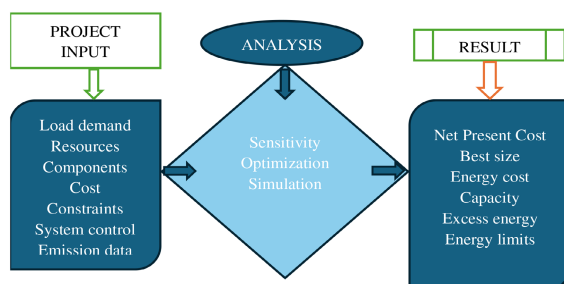


Figure 3. Result and discussion to solution implementation model.

The Homer generated results for the best-case scenario for Least Net Present cost (NPC) = Rwandan franc 3910000000, Initial Capital (IC) = Rwandan franc 117666000 Operation and Maintenance (O&M) = Rwandan franc 296623000/annum, Least Cost of Energy (LCOE) = Rwandan franc 9.54kWh, internal Rate of Return (IRR) = 38%, Return on investment (ROE) = 34%, Simple Payback Period (SPP) = 2.6 years is shown Table 4.

Table 2. Electrical summary (excess and unmet).

Component	Consumption (kWh/yr)	Percent
AC Primary Load	3,206,890	100
DC Primary Load	0	0
Deferrable Load	0	0
Grid Sales	75.9	0.00237
Total	3,206,966	100

The data retrieved according to the simulated results include the electrical production, Net Present Cost (NPC), and Annualized Cost, is shown in Tables 2 to 4, and electrical production and consumption is shown in Tables 5 and 6. Also, the grid rate, renewable energy summary, and compare economics is provided in Tables 7 to 9. The cost summary, cash flow charts, proposed annual nominal cash flows, base system annual nominal cash flows, and cumulative discounted cash flows of the Advanced Hybridge Renewable System (AHRES) is shown in Figures 4 to 8. This optimized value ensures reliability and continuous power supply of 250kWh (minimum energy threshold expected to increase to the average energy threshold of 500kWh) of modern and clean energy to the household with additional health and environmental gains, appealing payback period and low or net zero emissions. The sensitivity analyses shows that the variation in component parametric dimensions is within +/-10%, which is smooth and represents narrow wind speed variations, almost constant average sunshine during the day, and fairly water stream uniform velocity, with near zero acceleration. The hydrokinetics energy resource is not only affordable, but is made from simple technology, that can be transferred to the republic of Rwanda, for local skill content development in the power sector, and reducing the chances of importation of this vital component. This sensitivity includes wind speed, hydro speed, battery cost, sunshine

frequency (Solar radiant energy frequency must be greater than Threshold frequency (constant) + work function threshold).

Table 3. Net Present Costs (NPCs).

Name	Capital Cost	Operating Cost	Replacement Cost	Salvage Value	Resource Cost	Total Cost
10kW	\$80,000	\$34,193	\$0.00	\$0.00	\$0.00	\$114,193
Grid	\$0.00	\$4.14M	\$0.00	\$0.00	\$0.00	\$4.14M
Guinard-P154 [20kW]	\$37,666	\$42,741	\$11,541	-\$6,817	\$0.00	\$85,131
System	\$117,666	\$4.22M	\$11,541	-\$6,817	\$0.00	\$4.34M

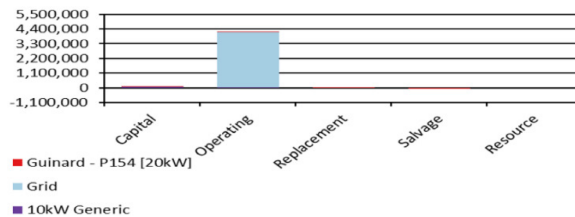


Figure 4. Cost summary.

Table 4. Annualized Costs (ACs).

Name	Capital	Operating	Replacement	Salvage	Resource	Total
10kW Generic	\$5,615.00	\$2,400.00	\$0.00	\$0.00	\$0.00	\$8,015.00
Grid	\$0.00	\$290,901.00	\$0.00	\$0.00	\$0.00	\$290,901.00
Guinard – P154	\$2,644.00	\$3,000.00	\$810.04	-\$478.46.00	\$0.00	\$5,975.00
System	\$8,259.00	\$296,301.00	\$810.04	-\$478.46.00	\$0.00	\$304,892.00

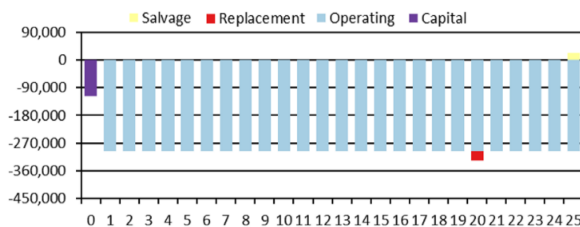


Figure 5. Cash flow chart.

Table 5. Electrical production summary.

Component	Production (kWh/yr)	Percent
Guinard – P154 [20kW]	175,200.00	5.46
Hydro	122,716.00	3.83
Grid Purchases	2,909,050.00	90.70
Total	3,206,966.00	100.00

Table 6. Electrical consumption summary.

Component	Consumption (kWh/yr)	Percent
AC Primary Load	3,206,890	100.00000
DC Primary Load	0.00	0.00000
Deferrable Load	0.00	0.00000
Grid Sales	75.90	0.00237
Total	3,206,966.00	100.00000

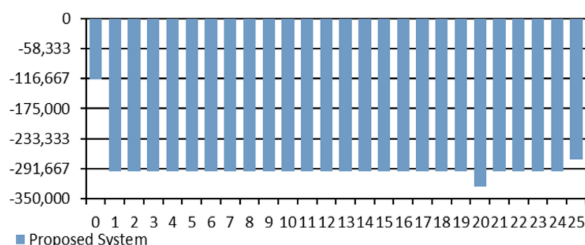


Figure 6. Proposed Annual Nominal Cash Flows (PANCF).

The BSANCF is \$340,000 after 20 years, and the PANCF is \$304,892 for 25 years.

Table 7. Grid rate: All.

Month	Energy Purchased (kWh)	Energy Sold (kWh)	Net Energy Purchased (kWh)	Peak Demand (kW)	Demand Charge	Energy Charge
January	243,967.00	6.1100	243,961	1,471	\$24,396.00	\$0.00
February	217,132.00	123.0000	217,118	1,281	\$21,713.00	\$0.00
March	253,777.00	0.0000	253,777	1,448	\$25,378.00	\$0.00
April	24,001,625.00	3.0000	241,622	1,480	\$24,162.00	\$0.00
May	240,978.00	15.2000	240,963	1,328	\$24,097.00	\$0.00
June	241,197.00	0.0000	241,197	1,236	\$24,120.00	\$0.00
July	244,985.00	0.0000	244,985	1,524	\$24,499.00	\$0.00
August	256,447.00	0.0000	256,447	1,416	\$25,645.00	\$0.00
September	243,087.00	0.0588	243,087	1,424	\$24,309.00	\$0.00
October	241,976.00	5.3500	241,970	1,360	\$24,197.00	\$0.00
November	236,966.00	32.9000	236,933	1,597	\$23,695.00	\$0.00
December	246,913.00	0.0000	246,913	1,326	\$24,691.00	\$0.00
Annual	2,909,050.00	75.9000	2,908,974	1,597	\$290,901.00	\$0.00

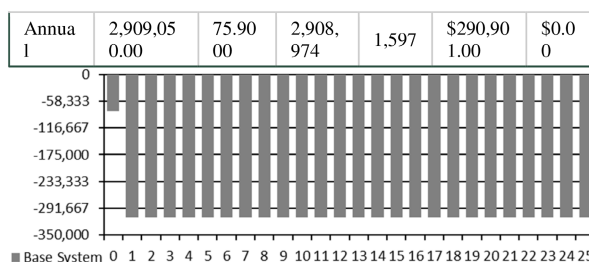


Figure 7. Base System Annual Nominal Cash Flows (BSANCF).

Table 8. Renewable summary.

Capacity-based metrics	Value	Unit
Nominal renewable capacity divided by total nominal capacity	100.00	%
Usable renewable capacity divided by total capacity	100.00	%
Energy-based metrics	Value	Unit
Total renewable production divided by load	9.29	%
Total renewable production divided by generation	9.29	%
One minus total nonrenewable production divided by load	100.00	%
Peak values	Value	Unit
Renewable output divided by load (HOMER standard)	82.40	%
Renewable output divided by total generation	100.00	%
One minus nonrenewable output divided by total load	100.00	%

Table 9. Compare economics.

IRR (%):38.5	Discounted payback (yr):2.85	Simple payback (yr):2.5
	Base System	Proposed System
Net Present Cost	\$4.51M	\$4.34M
CAPEX	\$80,000.00	\$117,666.00
OPEX	\$310,817.00	\$296,633.00
LCOE (per kWh)	\$0.0987.00	\$0.0951.00
CO ₂ Emitted (kg/yr)	1,949,198.00	1,838,520.00
Fuel Consumption (L/yr)	0.00	0.00

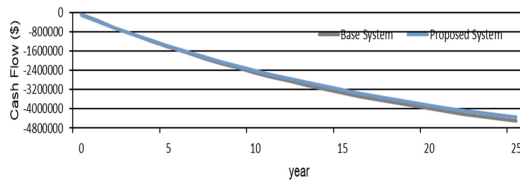


Figure 8. Cumulative discounted cash flows.

The top 3 system configurations, as determined by Homer, in a comparative analysis, and the homer software simulated results are clearly depicted in Figures 4 to 8. The top three is different due to different component combinations, but they are net zero emitters, except the national grid power. Another difference is in the initial cost capital ratio, which is 118: 119: 125. Thus, the first two configuration is better than the last third component configuration. The HOMER pro ensures that gas powered power plant is only used after all the renewable sources are fully engaged. Tables 3 to 9 shows that both the discounted and simple payback period is within 3 years, the compare economics for base and proposed system are similar, the discounted cash flow decreases with respect to time and spans across 25 years durations, the electrical consumption is 3,206,966 kWh/annum, and the system annualized cost is \$304,892. The wind energy speed according to Table 1 is above 2.0 m/s but less than 2.5 m/s from April to September yearly and lower than 2.0 m/s in the other months. This coupled with uneven hills in Rwanda, makes wind energy unattractive in Rwanda. This is also shown in Table 2, depicting wind energy capital cost and operating cost of \$80,000 and \$34,193, giving a total cost of \$114,193. This is higher that the capital, operating and total cost of other sources of renewable energy like the solar and hydro energy sources, yet these renewable energy sources have a low sensitivity, more reliable and higher yield in Rwanda than the wind energy sources.

8 CONCLUSION AND RECOMMENDATIONS

In the first case scenario, the Homer Pro recommended components are hydrokinetics, hydro-energy, and battery energy sources. This complies with the International Renewable Energy Agency (IREA) requirements, that recommends the selection of primary energy sources in the community, with the best renewable energy source potential. The second option presents all the renewable sources except wind energy sources, but at a high cost. Hence considering the low and unpredictable wind turbine output, it is not the optimum choice base on sensitivity analysis. The third choice, has a solar PV component, but based on optimization and sensitivity analysis, and considering the NPC and COE, wind energy sources is not recommended. The first two cases are different in terms of components implemented and initial cost of investment, which is a difference of 1,021,877.00 Rwandan franc. Also, the last two is within the same amount in terms of initial cost of investment but differ from the last two by almost 10,218,770.00 Rwandan franc. Hence, the first two component configuration is much affordable than the last two component configurations. The importance of the Homer results and the impact they have for the affected community is reflected in the revelation that wind energy source is not adequate and accessible. Indeed, in Cyangugu, the solar irradiation is available and reliable based on HOMER Pro results and offers the best energy source with the least carbon footprint. This is followed by the hydro resources, and the least renewable energy source is the wind because of the numerous hills blocking the wind. Thus, the result is a good reflection of what is best for this rural community, and it truly agrees with all the researched articles cited in this paper. These results will have a positive impact on this remote community of Rwanda because it assesses the energy demand of the vulnerable and poor households and proffer sustainable and reliable solution, with the least effect on the climate, environment, GHGs, and other far-reaching effects.

The quality of life, air quality, environment, and health of this remote community will be improved if this erudite and meticulously researched component configuration integrated with microgrid controller and national grid is implemented. Either first or second case should be implemented for increased energy access that is above the energy poverty line (250kWh/annum) at an affordable rate. Also, the time spent in fetching firewood, and the inconveniences and health problems associated with cooking with firewood and biomass will be mitigated, since 80% of Rwandans use Biomass as a primary energy source. Finally, economics, emissions, renewable energy penetration, Levelized COE, operating cost, HOMER cycle charging, energy-based metrics, capacity-based metrics, peak values, cash flow, cash summary, electrical performance, and efficiency is done by HOMER-Pro since conversion efficiency and high.

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